Verify Single-Area OSPFv2

2.6.1

Verify OSPF Neighbors

If you have configured single-area OSPFv2, you will need to verify your configurations. This topic details the many commands that you can use to verify OSPF.

As you know, the following two commands are particularly useful for verifying routing:

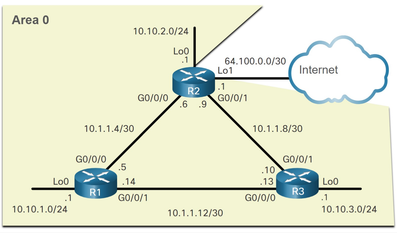
- **show ip interface brief** – This verifies that the desired interfaces are active with correct IP addressing.
- **show ip route**– This verifies that the routing table contains all the expected routes.

Additional commands for determining that OSPF is operating as expected include the following:

- **show ip ospf neighbor**
- **show ip protocols**
- **show ip ospf**
- **show ip ospf interface**

The figure shows the OSPF reference topology used to demonstrate these commands.

OSPFv2 Reference Topology



Use the **show ip ospf neighbor** command to verify that the router has formed an adjacency with its neighboring routers. If the router ID of the neighboring router is not displayed, or if it does not show as being in a state of FULL, the two routers have not formed an OSPFv2 adjacency.

If two routers do not establish adjacency, link-state information is not exchanged. Incomplete LSDBs can cause inaccurate SPF trees and routing tables. Routes to destination networks may not exist, or may not be the most optimum path.

Note: A non-DR or BDR router that has a neighbor relationship with another non-DR or BDR router will display a two-way adjacency instead of full.

The following command output displays the neighbor table of R1.

```
R1# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           0   FULL/  -        00:00:39    10.1.1.13       GigabitEthernet0/0/1
2.2.2.2           0   FULL/  -        00:00:38    10.1.1.6        GigabitEthernet0/0/0
R1#
```

For each neighbor, this command displays the following:

- **Neighbor ID** – This is the router ID of the neighboring router.
- **Pri** – This is the OSPFv2 priority of the interface. This value is used in the DR and BDR election.
- **State** – This is the OSPFv2 state of the interface. FULL state means that the router and its neighbor have identical OSPFv2 LSDBs. On multiaccess networks, such as Ethernet, two routers that are adjacent may have their states displayed as 2WAY. The dash indicates that no DR or BDR is required because of the network type.
- **Dead Time** – This is the amount of time remaining that the router waits to receive an OSPFv2 Hello packet from the neighbor before declaring the neighbor down. This value is reset when the interface receives a Hello packet.
- **Address** – This is the IPv4 address of the interface of the neighbor to which this router is directly connected.
- **Interface** – This is the interface on which this router has formed adjacency with the neighbor.

Two routers may not form an OSPFv2 adjacency if the following occurs:

- The subnet masks do not match, causing the routers to be on separate networks.
- The OSPFv2 Hello or Dead Timers do not match.
- The OSPFv2 Network Types do not match.
- There is a missing or incorrect OSPFv2 network command.

2.6.2

Verify OSPF Protocol Settings

The **show ip protocols** command is a quick way to verify vital OSPF configuration information, as shown in the following command output. This includes the OSPFv2 process ID, the router ID, interfaces explicitly configured to advertise OSPF routes, the neighbors the router is receiving updates from, and the default administrative distance, which is 110 for OSPF.

```
R1# show ip protocols
*** IP Routing is NSF aware ***
(output omitted)
Routing Protocol is "ospf 10"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
  Routing on Interfaces Configured Explicitly (Area 0):
    Loopback0
    GigabitEthernet0/0/1
    GigabitEthernet0/0/0
  Routing Information Sources:
    Gateway         Distance      Last Update
    3.3.3.3              110      00:09:30
    2.2.2.2              110      00:09:58
  Distance: (default is 110)
R1#
```

2.6.3

Verify OSPF Process Information

The **show ip ospf** command can also be used to examine the OSPFv2 process ID and router ID, as shown in the following command output. This command displays the OSPFv2 area information and the last time the SPF algorithm was executed.

```
R1# show ip ospf
  Routing Process "ospf 10" with ID 1.1.1.1
  Start time: 00:01:47.399, Time elapsed: 00:12:32.320
  Supports only single TOS(TOS0) route
  Supports opaque LSA
  Supports Link-local Signaling (LLS)
  Supports area transit capability
  Supports NSSA (compatible with RFC 3101)
  Supports Database Exchange Summary List Optimization (RFC 5243)
  Event-log enabled, Maximum number of events: 1000, Mode: cyclic
  Router is not originating router-LSAs with maximum metric
  Initial SPF schedule delay 5000 msecs
  Minimum hold time between two consecutive SPFs 10000 msecs
  Maximum wait time between two consecutive SPFs 10000 msecs
  Incremental-SPF disabled
  Minimum LSA interval 5 secs
  Minimum LSA arrival 1000 msecs
  LSA group pacing timer 240 secs
  Interface flood pacing timer 33 msecs
  Retransmission pacing timer 66 msecs
  EXCHANGE/LOADING adjacency limit: initial 300, process maximum 300
  Number of external LSA 1. Checksum Sum 0x0041FF
  Number of opaque AS LSA 0. Checksum Sum 0x000000
  Number of DCbitless external and opaque AS LSA 0
  Number of DoNotAge external and opaque AS LSA 0
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Number of areas transit capable is 0
  External flood list length 0
  IETF NSF helper support enabled
  Cisco NSF helper support enabled
  Reference bandwidth unit is 10000 mbps
    Area BACKBONE(0)
        Number of interfaces in this area is 3
        Area has no authentication
        SPF algorithm last executed 00:11:31.231 ago
        SPF algorithm executed 4 times
        Area ranges are
        Number of LSA 3. Checksum Sum 0x0E077E
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
R1#
```

2.6.4

Verify OSPF Interface Settings

The **show ip ospf interface** command provides a detailed list for every OSPFv2-enabled interface. Specify an interface to display the settings of just that interface, as shown in the following output for Gigabit Ethernet 0/0/0. This command shows the process ID, the local router ID, the type of network, OSPF cost, DR and BDR information on multiaccess links (not shown), and adjacent neighbors.

```
R1# show ip ospf interface GigabitEthernet 0/0/0
GigabitEthernet0/0/0 is up, line protocol is up
  Internet Address 10.1.1.5/30, Area 0, Attached via Interface Enable
  Process ID 10, Router ID 1.1.1.1, Network Type POINT_TO_POINT, Cost: 10
  Topology-MTID    Cost    Disabled    Shutdown      Topology Name
        0           10        no          no            Base
  Enabled by interface config, including secondary ip addresses
  Transmit Delay is 1 sec, State POINT_TO_POINT
  Timer intervals configured, Hello 5, Dead 20, Wait 20, Retransmit 5
    oob-resync timeout 40
    Hello due in 00:00:01
  Supports Link-local Signaling (LLS)
  Cisco NSF helper support enabled
  IETF NSF helper support enabled
  Index 1/2/2, flood queue length 0
  Next 0x0(0)/0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
R1#
```

To get a quick summary of OSPFv2-enabled interfaces, use the **show ip ospf interface brief** command, as shown in the following command output. This command is useful for seeing important information including the following:

- Interfaces are participating in OSPF
- Networks that are being advertised (IP Address/Mask)
- Cost of each link
- Network state
- Number of neighbors on each link

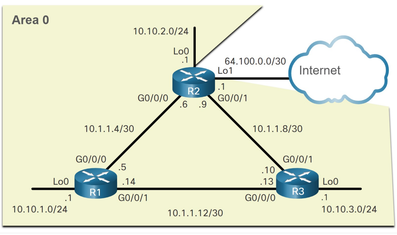
```
R1# show ip ospf interface brief
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo0          10    0               10.10.1.1/24       10    P2P   0/0
Gi0/0/1      10    0               10.1.1.14/30       10    P2P   1/1
Gi0/0/0      10    0               10.1.1.5/30        10    P2P   1/1
R1#
```

2.6.5

Syntax Checker – Verify Single-Area OSPFv2

Use the Syntax Checker to verify single-area OSPFv2 configuration on R2 and R3.

OSPF Reference Topology




```
R2# show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   10.1.1.6        YES NVRAM  up                    up
GigabitEthernet0/0/1   10.1.1.9        YES NVRAM  up                    up
Serial0/1/0            unassigned      YES NVRAM  administratively down down
Serial0/1/1            unassigned      YES NVRAM  administratively down down
GigabitEthernet0       unassigned      YES NVRAM  administratively down down
Loopback0              10.10.2.1       YES NVRAM  up                    up
Loopback1              64.100.0.1      YES NVRAM  up                    up
R2#

Display the OSPF routes installed in the routing table on R2.

R2#
```

Reset Show Me Show All

2.6.6

Packet Tracer – Verify Single-Area OSPFv2

In this Packet Tracer activity, you will use a variety of commands to verify the single-area OSPFv2 configuration.

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